SUIDE COUNTY URBAN MSW LANDFILL BIOGAS POLLUTION CONTROL AND COMPREHENSIVE UTILIZATION PROJECT

Report ID	*A+SH_SYST_VCS_VAL_5624*
Project title	*Suide County Urban MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project*
Project ID	*5281*
Crediting period	*24/02/2023 to 23/02/2033 (both days included)*
Original date of issue	*28/07/2025*
Most recent date of issue	*28/07/2025*
Version	*01.1*
VCS Standard Version	*VCS Standard version 4.7*
Client	*Henan BCCY Environmental Energy Co., Ltd.*
Prepared by	*LGAI Technological Center, S.A. (Applus+ Certification)*

Approved by	*LGAI Technological Center S.A. (Applus+ Certification)* *VVB Technical Manager: Mr. Agustín Calle de Miguel*
Work carried out by	*Lead Auditor / Technical Expert: Doris Dai* *Technical Reviewer: Simon Shen*

Summary:

LGAI Technological Center, S.A. (hereafter referred to as "Applus+ Certification") has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of project activity "Suide County Urban MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project" (hereafter referred to as "the project activity") and reported in the PD.

The project activity is a landfill gas recovery and power generation project located at Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, P. R. China, of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 1 MW consisting of 2 sets of 500 kW generators. The project uses LFG from Suide landfill site for power generation. During the 10-year fixed crediting period, the power generated by the project is expected to be 57,844 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of North China Power Grid (NWCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the emission of methane that would be generated under landfill condition. It's estimated that the project activity could achieve GHG emission reductions of 352,588 tCO_2e during the 10-year fixed crediting period.

The project activity is available at https://registry.verra.org/app/projectDetail/VCS/5281.

In the course of Validation, 00 Corrective Action Requests (CARs), 01 Clarification Requests (CLs) and no Forward action request (FARs) were raised.

The objective of this validation activity is to have an independent third party for the assessment of the project design, estimated ER sheet and to ensure a thorough assessment of the proposed project activity against the applicable CDM and VCS requirements. In particular;

- *AMS-III.G: Landfill methane recovery, version 10.0 /4/*
- *AMS-I.D: Grid connected renewable electricity generation, version 18.0 /4/*
- *Emissions from solid waste disposal sites, version 08.1 /8/*
- *Tool to calculate the emission factor for an electricity system, version 07.0 /7/*
- *Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0 /10/*
- *Positive lists of technologies, version 04.0 /11/*

The projects compliance with the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS Program Guide, version 4.4 /12/ and VCS Standard version 4.7 /13/.

- *CDM Validation and Verification Standard for project activities, version 03.0 /5/*
- *CDM Project Standard for project activities, version 03.0 /6/*
- *VCS Standard, version 4.7*
- *VCS Program Guide, version 4.4*

Validation is a requirement for all VCS projects and is seen as necessary to provide assurance to stakeholders of the quality of the project and its intended generation of estimated Verified Carbon Units (VCUs).

A risk-based approach has been followed to perform this validation activity. VVB applied a materiality threshold of 5% with respect to omission or misstatements concerning reported quantities as per VCS Standard version 4.7 clause 4.1.10 (4). The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews and project owners have provided LGAI Technological Center S.A. (Applus+ Certification) with sufficient evidence to verify the fulfilment of the stated criteria of VCS.

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1 INTRODUCTION

1.1 Objective

LGAI Technological Center, S.A. (hereafter referred to as "Applus+ Certification") has been commissioned by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of project activity "Suide County Urban MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project" (hereafter referred to as "the project activity") and reported in the PD.

LGAI Technological Center, S.A. as the validation and verification body of the project activity has been accredited as a DOE by UNFCCC and also meets the competence requirements as set out in ISO 14065: 2020.

The objective of this validation is to ensure that reported information in the PD of "Suide County Urban MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project" is complete and accurate in accordance with applicable VCS standards and relevant UNFCCC requirements.

1.2 Scope and Criteria

The validation scope is defined as an independent and objective review of the project design (PD). The PD is reviewed against the criteria stated in Article 12 of the Kyoto Protocol, the CDM modalities and procedures as agreed in the Marrakech Accords and the relevant decisions by the CDM Executive Board, including the approved baseline and monitoring methodology AMS-III.G version 10.0 and AMS-I.D version 18.0. The validation was based on the requirements in the CDM Validation and Verification standard for project activities version 03.0 and VCS program guide version 4.4 and standard version 4.7.

The assessment team has employed a risk-based approach to assess the completeness and accuracy of the claims and conservativeness of the assumptions in the VCS PD. The main focus of the assessment team is to identify the significant risks for the project implementation and the generation of VCUs. The validation is not meant to provide any consulting towards the project proponent. However, stated requests for clarifications and/or corrective actions may have provided input for improvement of the project design.

The only purpose of the validation is its usage during the registration/issuance processes as part of the VCS project cycle. Therefore, LGAI Technological Center S.A. (Applus+ Certification) can't be held liable by any party for decisions made or not made based on the validation opinion, which will go beyond that purpose.

1.3 Reasonableness of Assumptions

Applus+ Certification takes into consideration that the risk of material misstatements as part of the performance of a VCS validation assessment varies with factor/elements such as (i) the complexity and subjectivity associated with the process; (ii) the availability and reliability of relevant data; (iii) quality and quantity of information and evidences; (iv) the number and significance of assumptions that are made and finally, (v) the degree of uncertainty associated with such factors/elements.

In the context of the performance of the VCS validation assessment vis-à-vis considered assumptions, under conformance with Applus+ Certification working procedures, the validation team appointed by Applus+ Certification is required to perform an evaluation/test of the overall reasonableness, completeness and accuracy of all relevant assumptions used for the description of the design of the proposed VCS project activity, the project start date, the delineation of the project boundary, the identification of the baseline scenario, assessment and demonstration of additionality, applied GHG calculation approaches and the design of the monitoring plan (based on ex-ante determined parameters and parameters to be monitored ex-post).

Such reasonability analysis is performed by the Applus+ Certification's validation team (that holds significant experience and expertise with project-based initiatives encompassing collection and destruction/utilization of LFG) by assuming related limitations and by inter-alia taking into consideration the following relevant aspects:

- *Regional, national and international practice related to GHG mitigation/abatement measure and technology applied by the proposed VCS project activity.*
- *Definitions, concepts and principles of the applied baseline and monitoring methodology and methodological tools (which are assumed as being designed by being rooted in sound and widely accepted scientific principles and knowledge).*
- *Overall credibility of quality and reliability of data, information, facts and evidences used inter-alia for the identification of baseline scenario and baseline emissions (with consideration of contrary or inconsistent data, information and evidence where applicable).*
- *Relevance and credibility of identified stakeholders for the proposed VCS project activity.*
- *Regulatory, legal and operational requirements related to the measure and technologies encompassed by the proposed VCS project activity.*

Also, part of its validation assessment, a risk assessment is therefore performed as part of the validation planning in order to identify (and mitigate, if applicable) potential factors and/or elements that may negatively influence the credibility and fairness of the outcomes of the assessment. In case significant risks are identified, further tests may have been considered as required in order to evaluate the reasonableness of assumptions (and/or alternative assumptions) or outcomes, if applicable.

The validation report expresses a conclusion with a reasonable level of assurance about whether the reported GHG emissions reduction data is free from material misstatement. The assessment team applied a materiality threshold of 5% (for projects with ER less than or equal to 300,000 tCO_2e per year) with respect to emission or misstatements concerning reported quantities as per para 3.10.1 and 4.1.10 (4) of VCS Standard version 4.7.

Finally, it is also crucial to note that, as part of the technical review of the assessment, the reasonableness of the identification and eventual mitigation of such risks (as well as other relevant judgements and decisions made by the assessment team) is to be confirmed by the appointed review team (which is independent to the validation team).

In conclusion, Applus+ Certification planned and performed the Validation by obtaining objective evidence that substantiate the reasonableness of PP's assumptions and determine whether the project and its GHG statement conforms with the regulatory documents, as well as evaluates the limitations and methods that support a claim about the outcome of future activities, in accordance with the VCS Standard 4.7.

1.4 Summary Description of the Project

Project title	Suide County Urban MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project
VCS reference number	5281
Project Proponent	Henan BCCY Environmental Energy Co., Ltd. (Project Proponent, host country, P. R. China)
Location of the project	Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, P. R. China Geographic coordinates: Longitude: 110°15′41.788″ E, Latitude: 37°28′06.315″ N
Project start date	Construction date: 30/09/2022 Operation start date: 24/02/2023
Version of VCS PD	Version 03 dated 28/07/2025
Applied Methodology/Version	AMS-III.G, version 10.0 AMS-I.D, version 18.0
Scope/Technical Area	1/1.1 13/13.1

The project activity is a landfill gas recovery and power generation project located at Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi

Province, P. R. China, of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 1 MW consisting of 2 sets of 500 kW generators. The project uses LFG from Suide landfill site for power generation. During the 10-year fixed crediting period, the power generated by the project is expected to be 57,844 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of North China Power Grid (NWCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the emission of methane that would be generated under landfill condition. It's estimated that the project activity could achieve GHG emission reductions of 352,588 tCO_2e during the 10-year fixed crediting period.

2 VALIDATION PROCESS

2.1 Method and Criteria

The validation process has been conducted following the regulatory documents and based on the ISO 14064-3: 2019 and ISO 14065: 2020 using standard auditing techniques as required by the VCS Standard version 4.7. The validation process has been conducted through an onsite inspection (see below), thus, no need for a risk analysis for remote inspections has been determined for this Validation (cf. VCS Standard version 4.7. Para 4.1.13).

No ICT means have been used for the conduction of this validation process other than those for the provision of documents and transferring information and undertake normal communications with the PP(s). Those ICT means supporting the process in this sense, have been used to an extent sufficient to cover the background necessities of the validation process and have demonstrated enough effectiveness to conduct it in accordance with the relevant applicable regulatory requirements. No virtual sites have been included in the validation process according to the definition in the IAF MD4: 2023 document. Instead, the documentation has been provided upon request from the VVB by other means (such as physical documents, mail, transfer applications, etc.).

Applus+ Certification completed a strategic review and risk assessment of the project's activities and processes in order to gain a full understanding of (if applicable):

- Project Details;

- Application of Methodology;

- Estimated GHG Emission Reduction and Removals;

- Monitoring;

- Safeguards and SDG contributions etc.

- Activities associated with all the sources contributing to the project emissions and emission reductions, including leakage if relevant;

- Protocols used to estimate or measure GHG emissions from these sources;

- Collection and handling of data;

- Controls on the collection and handling of data;

Based on the above, the VVB prepared a Validation Plan and applied an evidence-gathering plan designed to collect sufficient and appropriate evidence upon which to base the conclusion and to consider the inherent risk for the validation. The evidence-gathering plan is based on analytical procedures and tests and considers the thresholds for materiality as per the regulatory documents applied for this validation (5% as per the applied VCS Standard version 4.7).

The applied evidence-gathering plan is used by the VVB to determine the conformity of the *statements made by the client against the principles and requirements of the applied regulatory documents during this validation. The evidence-gathering plan designed by the VVB takes into consideration, as a minimum:*

- The selection and management of the data and information;

- The processes for collecting, processing, consolidating, aggregating and reporting the data and information;

- Systems and processes in place to ensure the validity and accuracy of the data and information;

- The design and maintenance of the system to control the data and information;

- Systems, processes and personnel that support system in place, including activities for ensuring data quality;

- The plans for instrument maintenance and calibration where appropriate;

Applus+ Certification validate that the reported information in the Project Description is complete and accurate in question. This involved an on-site audit and a desk review of the Project Design. This Validation Report describes the findings of this assessment. The process of validation involved the following relevant milestones:

- The signature of the contract for validation between the PP and the VVB;

- A kick-off meeting/communication to require the documentary material to perform an initial desk review;

- An initial desk review of the documentary evidences prepared by the Project Proponent;

- The issuance of findings coming from this initial desk review (if any);

- The conduction of an onsite audit (see Sections 2.3 and 2.4 below);

- The issuance of findings from the onsite audit and communication of deviations (if any) to the Project Proponent;

- The assessment of their resolution and closure, when appropriate;

- Once all the findings have been closed, the elaboration of this Validation Report, in which such findings (if any) are described;

- The conduction of an independent Technical Review;

- Upon conclusion of the Technical Review, the VVB may perform a final quality check before the issuance of the Final Validation Report (FVR) and signature of the corresponding Deed.

The process of this validation assessment has not implied any sampling plan.

The information of the assessment team is included in below:

Assessment team

According to the sectoral scopes / technical area and experiences in the sectoral or national *business environment, LGAI Technological Center S.A. (Applus+ Certification) has composed a project assessment team in accordance with the appointment rules in the internal Quality Management System of LGAI Technological Center S.A. (Applus+ Certification). The composition of assessment team has to be approved by the LGAI Technological Center S.A. (Applus+ Certification) Central Site during its Contract Review stage, ensuring that the required skills are covered by the team. The qualification levels for Assessment Team members that are assigned by aforementioned appointment rules are as presented below:*

- Leader Auditor (LA)

- Auditor (A)/ Auditor Trainee (AT)

- Technical Reviewer (TR)

- Technical Experts (TE)

- Other supporting roles such as Country Expert (CE) or Financial Expert (FE)

- Any of the above-mentioned roles in training (iT, e.g. AiT for auditor in training).

The Sectoral Scope / Technical Area required knowledge linked to the applied methodology(ies) is covered by the assessment team as shown below:

Name	*Qualification*	*SS/TA Knowledge*	*Financial Expertise*	*Host country experience*	*Attendance to on-site visit*
Doris Dai	*LA/TE*	*Yes (1.1/13.1)*	*Yes*	*Yes*	*Yes*
Simon Shen	*TR*	*Yes (1/1/13.1)*	*Yes*	*n/a*	*n/a*

***Doris Dai** (Master's Degree in Environmental Sciences, Bachelor's Degree in Environmental Technology) is an Auditor appointed by Applus+ Certification (LGAI Technological Center, S.A) for the GHG project assessment and auditing. She has more than 8 years of work experience in CDM/VCS project assessment and more than 3 years of working experience in GS4GG project assessment. Before she joined Applus+ LGAI, she has been working for CTI Certification as senior GHG Auditor for 3.5 years.*

***Simon Shen** (Master's Degree in Thermal Energy Engineering, Bachelor's Degree in Environmental Engineering) is an Auditor appointed by Applus+ LGAI for the GHG project assessment, auditing and technical review. He has more than 8 years of work experience in CDM/GS4GG/VCS project assessment and review with Applus+, apart from the years of experience working as GHG Auditor and ISO 9001/14001 in TUV SUD for 3.5 years before he joined Applus+. Mr. Simon Shen has extensive experience also as former Applus+ Shanghai CDM Technical Manager.*

2.2 Document Review

The PD version 01 dated 16/11/2024, version 03 dated 28/07/2025 and the emission reduction calculations spreadsheet /2/, were assessed as part of the validation. Relevant documents were reviewed. A detailed documents reviewed are listed in Appendix II of the report.

2.3 Interviews

The key personnel interviewed are summarized in the table below:

Interviewed personnel	***Role***	***Organization***	***Subject***
Mr. Du Gaohui	*Manager*	*Suide County BCCY New Power CO., Ltd.*	*Status of the project (including PPs);* *Applicability of selected methodology;* *Baseline of the project and its updates;* *Emission factors and their updates;* *Monitoring plan;* *Stakeholder consultation process and its outcomes.* *The process and participation of the stakeholder consultation;* *The impact of the project activity;* *The complaint by local stakeholders and the implementation of the mitigation measures;* *Status of MSW landfill;* *Operation of MSW landfill.*
Mr. Liu Gang	*Manager*	*Suide landfill site*	
Mr. Xu Ming	*Staff*	*Suide landfill site*	
Ms. Xu Xiumin	*Villager*	*Shenjiawan Village*	
Ms. Zhao Gu	*Villager*	*Shenjiawan Village*	
Mr. Yang Weiming	*Villager*	*Shenjiawan Village*	
Mr. Zhou Xuetao	*Villager*	*Shenjiawan Village*	
Mr. Guo Jianjun	*Villager*	*Shenjiawan Village*	
Ms. Fang Yongmei	*Project Manager*	*Henan BCCY Environmental Energy Co., Ltd.*	*Data collection and ER calculation.*

2.4 Site Visits

The assessment team performed the on-site validation (Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, China) during 14/11/2024 - 15/11/2024. The interviewed personnel and objective are listed in above table.

2.5 Resolution of Findings

As an outcome of the validation process, the team can raise different types of findings.

Where a non-conformance arises the assessment team shall raise a Corrective Action Request (CAR). A CAR is issued, where:

a) Non-compliance with the monitoring plan or methodology are found in monitoring and reporting and has not been sufficiently documented by the Project proponents, or if the evidence provided to prove conformity is insufficient;

b) Modifications to the implementation, operation and monitoring of the project activity has not been sufficiently documented by the Project proponents;

c) Mistakes have been made in applying assumptions, data or calculations of emission reductions that will impact the quantity of emission reductions;

d) Issues identified in a FAR during validation to be verified during verification or previous verification(s) have not been resolved by the Project proponents.

The assessment team shall raise a Clarification Request (CL) if information is insufficient or not clear enough to determine whether the applicable CDM or VCS requirements have been met.

All CARs and CLs raised during validation shall be resolved prior to submitting a request for registration.

The objective of this phase of the validation was to resolve the requests for corrective actions and clarification and any other outstanding issues which need to be clarified for LGAI Technological Center S.A. (Applus+ Certification)'s positive conclusion on the PD. The Corrective Action Requests and Clarification Requests raised by LGAI Technological Center S.A. (Applus+ Certification) were resolved during communications between the Client and LGAI Technological Center S.A. (Applus+ Certification) to guarantee the transparency of the validation, the concerns raised, and responses given are summarized below in the Appendix III.

The PD version 03 dated 28/07/2025 serves as the basis for the final assessment presented. Additional changes to the project during the validation process are not considered to be significant with respect to the main CDM/VCS objectives. The two CDM/VCS main objectives are the reduction of anthropogenic GHG emissions and the contribution of sustainable development to the host country.

Areas of validation findings	***No. of CL***	***No. of CAR***	***No. of FAR***
Project Description and monitoring report	*00*	*00*	*00*
Description of project activity	*00*	*00*	*00*
Application of selected baseline and monitoring methodology and selected standardized baseline			
- Applicability of methodology and standardized baseline	*00*	*00*	*00*

- Deviation from methodology	*00*	*00*	*00*
- Clarification on applicability of methodology, tool and/or standardized baseline	*00*	*00*	*00*
- Demonstration of additionality	*01*	*00*	*00*
- Emission reductions	*00*	*00*	*00*
- Monitoring plan	*00*	*00*	*00*
- Stakeholders consultation process	*00*	*00*	*00*
- Environmental Impacts	*00*	*00*	*00*
- Public comments	*00*	*00*	*00*
Others (please specify)	*00*	*00*	*00*
Total	*01*	*00*	*00*

2.5.1 Forward Action Requests

No FAR was raised during the validation process.

3 VALIDATION FINDINGS

3.1 Project Details

The project activity is a landfill gas recovery and power generation project of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 1 MW consisting of 2 sets of 500 kW generators.

The project uses LFG from Suide landfill site for power generation. During the 10-year fixed crediting period, the power generated by the project is expected to be 57,844 MWh which will be exported to the grid. The Project can reduce GHG emissions by replacing the electricity generated by fossil fuel fired power plants of North China Power Grid (NWCPG). Meanwhile by utilization the landfill gas recovered by project, the project activity avoids the emission of methane that would be generated under landfill condition. It's estimated that the project activity could achieve GHG emission reductions of 352,588 tCO_2e during the 10-year fixed crediting period.

The project activity is located at Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, P. R. China. The project activity has been developed by Suide County BCCY New Power CO., Ltd. The geographic coordinates of the project activity are longitude of 110°15′41.788″ E and latitude of 37°28′06.315″ N which is confirmed by site visit.

The Project activity has been approved by Chinese government by checking the Project approval /16/ and Environmental Impact Assessment (EIA) approval /18/. The project started construction on 30/09/2022 confirmed by checking construction order /20/ and commissioned on 24/02/2023 by checking operation log /25/ and site visit.

Assessment team checked through on-site audit and confirms that the details of the project proponent is as below:

Organization name	Henan BCCY Environmental Energy Co., Ltd.
Contact person	Lei Wang
Title	Minister of carbon emission department
Address	10/F, Kailin Building, 99 Ruyi West Road, Zhengzhou, China
Telephone	+86 371 56737901
Email	lwang@bccynewpower.com

Assessment team checked through on-site audit and confirmed that the details of the other entity involved is as below:

Organization name	Suide County BCCY New Power CO., Ltd.

Role in the project	*Project Operation Entity*
Contact person	*Yongmei Fang*
Title	*Project Manager*
Address	*10/F, Kailin Building, 99 Ruyi West Road, Zhengzhou, China*
Telephone	*+86 371 56737901*
Email	*ymfang@bccynewpower.com*

By checking cooperation agreement between Urban Management Law Enforcement Bureau of Suide County and Henan BCCY Environmental Energy Co., Ltd. /26/ which is the mother company of Suide County BCCY New Power CO., Ltd. regarding to the utilization of landfill gas, the cooperation period for LFG generation will last after 23/02/2033.

Overall, it is confirmed that the PD is accurate, complete, and provides an understanding of the nature of the project.

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Audit history	*The validation of this project activity was conducted by LGAI Technological Center, S.A. (Applus+ Certification).*
Sectoral scope	*By checking PD and applied methodology, it is confirmed that the project falls under sectoral scope 01 and 13.*
AFOLU project category, if applicable	*Not applicable since the project activity was not an AFOLU project.*
Project activity type	*By checking PD and applied methodology, it is confirmed that the project activity type of the project belongs to Energy industries (renewable-/non-renewable sources) (scope 01) and Waste handling and disposal (scope 13).*
General eligibility of the project to participate in the VCS Program	*By checking VCS Standard Version 4.7, PD and site visit witness, it is confirmed that the project is excluded under Table 2.1 of the VCS Standard as a landfill gas power generation project.* *The project fully met the requirements related to pipeline listing deadline, the opening meeting with the VVB, and the validation deadline by checking information on the VCS website.*

	By checking VCS website, it is confirmed that the applied methodology for the project is eligible under the VCS Program as small-scale methodology. *By checking Project approval and site visit, it is confirmed that the project is not a fragmented part of a larger project or activity that would otherwise exceed such limits.* *No other relevant eligibility information detected during the validation process.*
AFOLU project eligibility, if applicable	*Not applicable since the project activity was not an AFOLU project.*
Transfer project eligibility, if applicable	*Not applicable since the project activity is not a Transfer project.*
Project design	*Not appliable since the project is not a grouped project activity.*
Project ownership	*Suide County BCCY New Power CO., Ltd. is the project owner of project activity and its mother company Henan BCCY Environmental Energy Co., Ltd. is project proponent (PP) of project activity, and they have the legal right to control and operate the project activity.* *The project ownership has been checked by the assessment team and demonstrated through checking business license /14/, Project Approval and EIA Approval.*
Project start date	*By checking operation log and via onsite interview, it was verified the project started commercial operation on 24/02/2023, which is the start date of the project activity and the date actual GHG mitigation activity started*
Project crediting period	*The project activity adopts 10-year fixed crediting period.* *The assessment team confirms that the crediting period dates for the project is as below:* *Crediting Period Start date: 24/02/2023* *Crediting Period End date: 23/02/2033*
Project scale	*Assessment team confirms that the project is a small-scale project under CDM scheme. The total installed capacity of the project activity is 1 MW.* *As the estimated annual average GHG emission reductions or removal per year is 35,258 tCO_2e which is less than 300,000*

	tonnes of CO_2e per year, thus the project falls in the category of Project under VCS Standard.
Likelihood of achieving estimated GHG emission reduction or removals	*As the estimated annual average GHG emission reductions or removal per year is 35,258 tCO_2e, which is less than 300,000 tCO_2e per year, thus the project falls in the category of Project.*

Year	***Estimated GHG emission reductions or removals (tCO_2e)***
24/02/2023 – 31/12/2023	*23,078*
2024	*29,216*
2025	*31,310*
2026	*33,370*
2027	*35,409*
2028	*37,418*
2029	*39,408*
2030	*41,381*
2031	*39,338*
2032	*37,399*
01/01/2033 – 23/02/2033	*5,261*
Total estimated ERs	*352,588*
Total number of crediting years	*10*
Average annual ERs	*35,258*

The above estimated emission reduction is confirmed by assessment team via emission reduction calculation spreadsheet. The calculation is conservative and this acceptable to the assessment team. The likelihood of achieving estimated GHG emission reduction is high.

Technologies and measures implemented by the project activity or activities	*The project activity is a landfill gas recovery and power generation project of which the main purpose is to use landfill gas for electricity generation. The total installed capacity of the project activity is 1 MW consisting of 2 sets of 500 kW generators.*

The project consists in LFG collection, transmission and treatment system, with subsequent electricity generation and grid connection system.

LFG collection system

LFG is extracted from MSW under negative pressure by blower pumps and moved through wells, then collected by gas collection stations and transferred by sub-pipes and a main pipe to LFG treatment equipment. Flow rate of the LFG is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

LFG treatment system

Prior to electricity generation, LFG is treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.

Electricity generation and grid connection system

3 gas engines of 500 kW rated power each are fed with the LFG and generate electricity, which is then exported to the grid.

The technical specifications are mentioned as below and the same were checked during on-site visit and checking technical specification of equipment /28/:

Main Equipment	***Parameter***	***Value***
Generator set	*Type*	*500GF-NK*
	Manufacturer	*CNPC JICHAI POWER EQIPMENT COMPANY*
	Rated capacity	*500 kW*
	Rated rotation speed	*1,000 r/min*
	Rated frequency	*50 Hz*
	Lifetime	*15 years*

By site visit interview, it is confirmed that the project does not include any flare system. As per ex ante estimation, the

	maximum operation power is 0.96 MW/h and the capacity of installed generator sets is 1 MW/h. It means that the generator sets can entire destroy the LFG generated by Suide landfill site. During the operation period, all LFG will be used to generate electricity and the LFG flow can be adjusted based on LFG demand as per operation power of the project. When the engines are scheduled to shut down, the LFG collection system will reduce the intake gas in advance. There is no LFG collected and left in the entire project system during the planned shutdown. The layout and length of the pipelines and other safety factors have been fully considered at the beginning of the design process, which will make the pipelines have the capacity to store the LFG for a period of time in case of emergency case. At the same time, the maintenance personnel will handle the malfunction as soon as possible. So, the LFG won't pass into the atmosphere.
Implementation schedule of the project activity or activities	*The Project activity has been approved by China's government by checking Project Approval and Environmental Impact Assessment (EIA) approval. The project started construction in 30/09/2022 confirmed by construction order and commissioned on 24/02/2023 by checking operation log and on-site visit interview.*
Project location	*The project activity is located at Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, China. The project activity has been developed by Suide County BCCY New Power CO., Ltd. The geographic coordinates of the project activity is longitude of 110°15′41.788″ E and latitude of 37°28′06.315″ N which is confirmed through checking online GPS map during site visit.*
Conditions prior to project initiation	*Before the implementation of the project activity, the electricity generated by the project would be supplied by NWCPG in the baseline scenario and the LFG combusted would be vented to atmosphere without utilization.* *The same has been confirmed by checking FSR, Project approval and site visit.*
Project compliance with applicable laws, statutes and other regulatory frameworks	*Assessment team confirms that the Project has been approved by P. R. China's government by checking the Project approval and Environmental Impact Assessment (EIA) approval.*

	By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.
Double counting and participation under other GHG programs	***Projects registered (or seeking registration) under other GHG program(s)*** *The project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. The declaration /29/ for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:* *- http://www.irecstandard.org/* *- http://cdm.unfccc.int/* *- http://www.goldstandard.org/* *- https://ccer.cets.org.cn/* *- http://verra.org/* ***Rejection by other GHG programs*** *The Project is not rejected by other GHG programs. A declaration for the same is checked and found correct by the assessment team. Also, assessment team checked the following registries to confirm the same. The details of the registries checked are as follows:* *- http://www.irecstandard.org/* *- http://cdm.unfccc.int/* *- http://www.goldstandard.org/* *- https://ccer.cets.org.cn/* *- http://verra.org/* *It is confirmed that the project has neither been registered nor seeking registration under any other GHG programs. The project is seeking registration only in VCS program. Applus+ Certification checked the REC Mechanism database of P. R. China and found that the project activity is not accredited /*

	registered under REC mechanism. Furthermore, although China has re-launch domestic GHG program as CCER, based on current regulation, the project is not eligible under CCER. Further, declaration for the same is checked and found correct by the assessment team. *The project has provided all information required on whether is registered or seeking registration under any other GHG programs as the project only apply under VCS Standard.* *The Project is not rejected by other GHG programs as the project only apply under VCS Standard.*
No double claiming with emissions trading programs or binding emission limits	*The assessment team confirms that the net GHG emission reductions or removals generated by the Project will not be used for compliance with an emissions trading program or to meet binding limits on GHG emissions in any Emission Trading program or other binding limits. The assessment team checked the REC Mechanism database of China and Chinese Emission Trading System (Chinese ETS) and found that the project activity is not accredited / registered under REC mechanism or Chinese ETS. Thus, the assessment team concluded that the project activity is not involved on other Emissions trading programs and other binding limits.* *Furthermore, as per "Notice on Key Work Related to the Management of Greenhouse Gas Emission Reports for Enterprises in 2022" /30/ issued by Ministry of Ecology and Environment of P. R. China, China has a national emissions trading scheme covering thermal power generation industry. However, power generation using landfill gas is excluded and thus no emission cap was enforced for the project proponent.*
No double claiming with other forms of environmental credit	*The Project has no intend to generate any other form of GHG-related environmental credit for GHG emission reductions or removals claimed under the VCS Program. Renewable energy certificates are available for trading in the host country. By checking REC Mechanism database of China /31/ as well as I-REC website /32/, it was verified only covering wind, solar, hydro and biomass power was covered. Therefore, it was verified the project activity is not included in REC Mechanism database of China.*

	In conclusion, the assessment team confirmed the project activity is not involved in other forms of environmental credit.
Supply chain (Scope 3) emissions double claiming	*It is confirmed that the project is to capture and utilize the landfill gas which otherwise will be released to atmosphere, and it will not impact the emissions of goods and services in a supply chain, i.e. Scope 3 emissions.*
Sustainable development contributions	*The project activity would contribute sustainable development in the region in following aspects confirmed by on-site audit:* ***SDG 7 Affordable and Clean Energy*** *The project utilizes the LFG from the Suide landfill site to generate electricity by gas engines. Total electricity generation supplied by the project to the grid is estimated to be 57,844 MWh, which could replace the equivalent electricity from fossil fuel based grid.* ***SDG 8 Decent works and Economic Growth*** *The project can provide full time job opportunities over the project lifetime.* ***SDG 13 Climate Action*** *The project utilizes the LFG for electricity generation, which contains nearly 50% of methane, to substitute the equivalent electricity supplied by the grid. Total GHG emission reductions and removals of 352,588 tCO_2e will be achieved during the fixed 10 years crediting period.* *Sustainable development priorities are illustrated in the People's Republic of China National Report on Sustainable Development issued on 01/06/2012. Sustainable development priorities of the Report related to the project are promoting new energy development, strengthening environmental pollution control and responding to climate change. Methane is an ideal clean fuel. Each cubic meter of methane contains about 360,000 kJ calorific value, LFG recovery and utilization will supplement the energy supply of the grid and contributes to the new energy development in China. Odour from landfill negatively impacts on residents around the landfill. Implementation of the project will reduce odours through LFG collection and will thus mitigate the impact*

	of landfill odour on people's daily lives. Landfill gas contains nearly 50% of methane and methane is a greenhouse gas some 28 times more potent than carbon dioxide. LFG recovery and utilization is contribution to mitigating global warming.

SDG Target	***SDG Indicator***	***Project activity contribution***
13.0	*Tonnes of greenhouse gas emissions avoided or removed*	*The project avoided anthropogenic emissions of greenhouse gases (GHG) of 352,588 tCO_2e during the crediting period (24/02/2023 to 23/02/2033).*
8.5	*Total number of job opportunities*	*The project activity employs 8 permanent and full-time employees through its operational company Suide County BCCY New Power CO., Ltd. during the crediting period (24/02/2023 to 23/02/2033).*
7.2	*MWh of renewable energy generated*	*Total 57,844 MWh of renewable energy generated during the crediting period (24/02/2023 to 23/02/2033).*

Additional information relevant to the project	***Leakage management for AFOLU projects***

	Not applicable to the project activity. ***Commercially sensitive information*** *No commercially sensitive information has been excluded from the public version of the project description. The details are presented transparently to the assessment team for analysis which lead to positive conclusion for this validation.* ***Any additional relevant information*** *As confirmed with PP, the assessment team is able to confirm that there are no additional relevant information that may have a bearing on the eligibility of the project, the reductions or removals, or the quantification of the project's reductions or removals.*

3.2 Safeguards and Stakeholder Engagement

3.2.1 Stakeholder Engagement and Consultation

3.2.1.1 Stakeholder Identification

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder identification	*By checking PD and site visit, the stakeholder involved in the project has been correctly identified and in line with the actual situation as below:* *The project is located in Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, P. R. China. local authority, employee of Urban Management Law Enforcement Bureau of Suide County (landfill site), employee of project and local residents are therefore considered as stakeholders. Surely the project proponent is also considered as stakeholder.* *The assessment team considers the stakeholder identification of the project activity as reasonable.*
Legal or customary tenure/access rights	*The project is located in Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, P. R. China.*

	By checking Cooperation agreement signed by the project owner and Urban Management Law Enforcement Bureau of Suide County, it is confirmed that the project owner has the legal right to use the LFG generated from Suide landfill site for power generation.
Stakeholder diversity and changes over time	*As stakeholders identified for this project activity are diverse in nature – in terms of socio-economic status, age and gender wise and the diversity is unlikely to change over time.*
Expected changes in well-being	*By site visit and checking Project approval, it is confirmed that expected changes in well-being as below:* *1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected;* *2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 57,844 MWh, which could replace the equivalent electricity from fossil fuel based grid.*
Location of stakeholders	*The project is located in Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, P. R. China, local authority, employee of Urban Management Law Enforcement Bureau of Suide County (landfill site), employee of project and local residents are therefore considered as stakeholders.* *The assessment team considers the stakeholder identification of the project activity as reasonable.*
Location of resources	*The location of resources is at Suide County MSW landfill site, Henjiawan Village, Mingzhou Town, Suide County, Yulin City, Shaanxi Province, P. R. China.*

3.2.1.2 Stakeholder Consultation and Ongoing Communication

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Stakeholder engagement process	*By checking PD, Stakeholder Meeting Notice /23/, Stakeholder Meeting Minutes /21/ and interviewing stakeholders and PP during site visit, it was verified a stakeholder meeting was held on 13/09/2022 to collect stakeholders' comments for the project*

	activity. Via interviewing stakeholders and PP during site visit, it was verified before the stakeholder consultation meeting, the project information with contact information were published on the bulletins at and nearby the project site on 29/08/2022 to invite Local stakeholders. *By checking questionnaires /22/, it was verified a survey was carried out to collect comments from local residents through distributing and collecting responses to a easy-designed questionnaire. PP considered the differences and interactions between the stakeholder groups, the stakeholders involved in the questionnaire survey were thereby local residents with different sex, age groups, education background and occupation, which are within different social, economic, and cultural diversity. The questionnaire includes a brief summary of the project activity, including the project design and other relevant key information. In total 15 out of 15 questionnaires were returned with a 100% response rate.*
Consultation outcome	*By checking questionnaires, meeting minute and site visit, it is confirmed that during the stakeholder consultancy meeting, some stakeholders worried that the project will bring noise, land occupying and pollution on employees and local residents' living. For these issues, the project owner will offer some measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents. All attendances to the meeting were satisfied to the measures.*
Ongoing communication	*By checking grievance expression book and site visit, it is confirmed that expect for introducing the project information, grievance mechanism was also explained during the local stakeholder consultation meeting. A grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project Proponent will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Project Proponent will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible. Also,*

	the project proponent will regularly hold stakeholder meetings to exchange project progress, issues, and solutions with local residents. For employees of the enterprise, the project proponent establishes a labor union organization and holds at least one employee meeting every year to pay attention to the needs of each employee and collect their opinions to better operate the project and improve employee happiness. *Therefore, it is able to confirm the ongoing communication process with the local stakeholders and grievance handling procedure is in place with effectiveness.*
Stakeholder input	*By checking the questionnaires, meeting minutes and site visit, Applus+ Certification confirm that the local stakeholder has no major negative comments for the construction of the project activity, and minor negative comments have been considered with mitigation measurements.*

3.2.1.3 Free, Prior, and Informed Consent

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Obtaining consent	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government.* *By checking Cooperation agreement signed by the project owner and Urban Management Law Enforcement Bureau of Suide County, it is confirmed that the project owner has the legal right to use the LFG generated from Suide landfill site for power generation.* *By checking questionnaires, grievance expression book, meeting minutes and site visit, it is confirmed that local stakeholders held full right for FPIC.* *Therefore, it was verified the project activity does not violate any legal rights held by stakeholders, indigenous people (IPs), local communities (LCs), and customary rights holders.*
Outcome of FPIC discussion	*As mentioned above, the project has been approved by Chinese government and local management institute. And local stakeholders have no major negative comments for the construction and operation of the project.*

	The project has not encroached on land, relocated people without consent, and forced physical or economic displacement.

3.2.1.4 Grievance Redress Procedure

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Development process	*By checking grievance expression book /24/ and site visit, it is confirmed that a grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project proponent will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Also, the project proponent will regularly hold stakeholder meetings to exchange project progress, issues, and solutions with local residents. For employees of the enterprise, the project proponent establishes a labor union organization and holds at least one employee meeting every year to pay attention to the needs of each employee and collect their opinions to better operate the project and improve employee happiness.* *Therefore, it is able to confirm the ongoing communication process with the local stakeholders and grievance handling procedure is in place with effectiveness.*
Grievance redress procedure	*By checking grievance expression book and site visit, it is confirmed that a grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project proponent will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Also, the project proponent will regularly hold stakeholder meetings to exchange project progress, issues, and solutions with local residents. For employees of the enterprise, the project proponent establishes a labor union organization and holds at least one employee meeting every year to pay attention to the needs of each employee and collect their opinions to better operate the project and improve employee happiness.* *Therefore, it is able to confirm the ongoing communication process with the local stakeholders and grievance handling procedure is in place with effectiveness.*

3.2.1.5 Public Comments

Comments received	Actions taken by the project proponent	Evidence gathering activities, evidence checked, and assessment conclusion
The assessment team noted that the project activity was open for public comment from 09/10/2024 to 08/11/2024. The detail was checked by the assessment team in the following web platform: https://registry.verra.org/app/projectDetail/VCS/5281. *During this period, no public comments were received.*	*Not applicable since no public comments were received during public comment period.*	*Not applicable since no public comments were received during public comment period.*

3.2.2 Risks to Local Stakeholders and the Environment

3.2.2.1 Management Experience

By checking Business License, it was verified the PP was established on 06/01/2022 with main business scope containing "power generation". Therefore, the assessment team deemed PP has expertise or experience in implementing similar project activities and engaging communities.

By checking EIA Approval and Project Approval issued by government, it is confirmed the project has been approved by Chinese government which means the government recognize the experience of PP for project construction and operation. Moreover, the project owner of the project has developed several LFG projects and therefore has extensive development and management experience.

3.2.2.2 Risk Assessment

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Natural and human-induced risks to stakeholders' wellbeing	*By checking EIA Approval and Project Approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project meets all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety.* *On the other hand, the project would contribute following changes in well being according to project approval:*

	1. Economic well-being – the project activity is likely to provide employments to the locality thus economic well beings are expected; *2. Environmental well-being – (1) as this project activity reduces the methane emissions in atmosphere by recovering and utilizing LFG to generate electricity, it will bring environmental well-being in the locality. (2) Total electricity generation supplied by the project to the grid is estimated to be 57,844 MWh, which could replace the equivalent electricity from fossil fuel based grid.* *Therefore, it was verified no natural and human-induced risks has been identified to stakeholders' wellbeing.*
Risks to stakeholder participation	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety.* *Also, during the site visit, no risks has been identified to stakeholder participation.* *Therefore, it was verified no risks has been identified to stakeholder participation.*
Working conditions	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety.* *Also, during the site visit, it is confirmed that the staff of the project is working under indoor environment and safety protection equipment have been provided in case for on-site monitoring and device maintenance. No risks identified for working conditions.*
Safety of women and girls	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. Moreover, China has published "Women's Rights Protection*

	Law" /33/ to provide support and protection to women and girls. *In order to protect the physical and mental health of minors, the State Council promulgated the "Prohibition of Child Labor Regulations" /33/, prohibiting employers from recruiting minors under the age of 16. Also, China published "Women's Rights Protection Law". Therefore, the Project will not expose women and girls to further risks or hazards. By checking HR Records and Staff Roaster /34/, it was verified no girls have been recruited, also equal payment would be made to women and men in case.* *Also, during the site visit, it is confirmed that no risks has been identified to safety of women and girls.*
Safety of minority and marginalized groups, including children	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of minor protection.* *The country has strict prohibition for child labour. Thus, project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview.* *Besides, in order to protect the physical and mental health of minors, the State Council promulgated the "Prohibition of Child Labor Regulations", prohibiting employers from recruiting minors under the age of 16. By checking HR Records and Staff Roaster and site interview, the assessment team was able to confirmed that no Minors under the age of 16 were employed.* *Also, during the site visit, it is confirmed that no minors have been hired by the project owner. No child labor detected for the project. No risks identified for safety of minority and marginalized groups, including children as not applicable for the project.*
Pollutants (air, noise, discharges to water, generation	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by*

of waste, release of hazardous materials, and chemical pesticides and fertilizers)	*Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of environment, health and safety. During construction phase, the raise dust, noise, wastewater and solid wastes caused by project construction will be treated according to the measures in EIA report /18/, and there will be no significant impact on the environment.* *During the operation period, all the mitigation measures proposed by EIA report will be implemented and the following key aspects will be addressed:* ***Waste Gas:*** *The main source of exhaust gas pollution in this project is the exhaust gas from the generator set. Each generator set is equipped with a 15m high exhaust funnel for high-altitude emissions. The exhaust gas can meet the limits listed in Table 2 of "Integrated Emission Standard of Air Pollutants" (GB16297-1996). It will not have a significant impact on the surrounding environment.* ***Waste water:*** *The waste water of the project is mainly condensate and domestic wastewater. After pre-treatment in the septic tank, the supernatant of domestic sewage is transported to the Suide County Sewage Treatment Plant for further treatment, and the manure residue is regularly cleaned. The condensate water is collected and transported through pipelines to the landfill leachate system for advanced treatment and finally used for watering in the landfill area. The project won't have a negative impact on the natural or pre-existing pattern of watercourses, ground-water and/or the watershed(s).* ***Noise pollution:*** *The noise is mainly generated by the generator set, pump equipment and other equipment. Through the adoption of low noise equipment, foundation shock absorption, building sound insulation, muffler, equipment daily maintenance and other measures, the noise contribution value of the factory boundary can meet the requirements of class 2 of Industrial Enterprise factory boundary Environmental noise Emission*

	Standard (GB12348-2008). The impact of production noise on the surrounding environment is relatively small. ***Solid waste:*** *The solid waste of the project is mainly general solid waste, waste filtering materials, waste engine oil and waste oil drum. The general solid waste is collected and transported to landfill site for landfilling. The waste filtering materials are belonging to general industrial solid waste, which will regularly collected and transported to the landfill site for landfilling. The waste engine oil and waste oil drum are belonging to hazardous waste, which will be stored in the hazardous waste room of the factory, and regularly handed over to qualified units for recycling and processing. The solid waste will not have a significant impact on the surrounding environment.*

3.2.3 Respect for Human Rights and Equity

3.2.3.1 Labor and Work

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Discrimination	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" /33/ and "Women's Rights Protection Law", which could prohibit such phenomena.* *By site visit, it is confirmed no discrimination and sexual harassment detected.* *The project does not create any direct or indirect impacts on gender equality and/or the situation of women:* *a. Minors under the age of 16 are not allowed to be recruited. Project participation by women or girls older than 16 is merely on voluntary basis and there is no compulsion on them. The project developer has a grievance cell which would investigate complaints. Through the site interview and checking HR Records and Staff Roaster, the assessment team was able to confirmed that no minors under the age of 16 were employed, women or girls older than 16 were employed on a voluntary basis, and no slavery, imprisonment, physical and mental drudgery, punishment or coercion to women and girls.*

	b. The Project will not restrict women's rights or access regarding natural resources. *c. Marital status is completely irrelevant to the Project. The project developer does not discriminate on gender, caste, religion etc.*
Sexual harassment	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena.* *By site visit, it is confirmed no discrimination and sexual harassment detected.* *The project does not create any direct or indirect impacts on gender equality and/or the situation of women:* *a. Minors under the age of 16 are not allowed to be recruited. Project participation by women or girls older than 16 is merely on voluntary basis and there is no compulsion on them. The project developer has a grievance cell which would investigate complaints. Through the site interview and checking HR Records and Staff Roaster, the assessment team was able to confirmed that no minors under the age of 16 were employed, women or girls older than 16 were employed on a voluntary basis, and no slavery, imprisonment, physical and mental drudgery, punishment or coercion to women and girls.* *b. The Project will not restrict women's rights or access regarding natural resources.* *c. Marital status is completely irrelevant to the Project. The project developer does not discriminate on gender, caste, religion etc.* *Therefore, it is confirmed equal opportunities have been provided by the project activity in the context of gender equity and pay for labor and work.*
Equal pay for equal work	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena.* *By site visit, it is confirmed no discrimination and sexual harassment detected.*

	The project does not create any direct or indirect impacts on gender equality and/or the situation of women: *a. Minors under the age of 16 are not allowed to be recruited. Project participation by women or girls older than 16 is merely on voluntary basis and there is no compulsion on them. The project developer has a grievance cell which would investigate complaints. Through the site interview and checking HR Records and Staff Roaster, the assessment team was able to confirmed that no minors under the age of 16 were employed, women or girls older than 16 were employed on a voluntary basis, and no slavery, imprisonment, physical and mental drudgery, punishment or coercion to women and girls.* *b. The Project will not restrict women's rights or access regarding natural resources.* *c. Marital status is completely irrelevant to the Project. The project developer does not discriminate on gender, caste, religion etc.* *Therefore, it is confirmed equal opportunities have been provided by the project activity in the context of gender equity and pay for labor and work.*
Gender equity in labor and work	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Women's Rights Protection Law", which could prohibit such phenomena.* *By site visit, it is confirmed no Gender equity in labor and work detected.* *The project, complying with the "Women's Rights Protection Law" and "Labor Contract Law" does not discriminate the local community on basis of gender or caste or religion and therefore equally serve to all.* *By checking HR Records and interviewing staff from the project owner, it was verified the Project design neither increase women's workload nor prevent them from engaging in other activities. The Project has equal opportunity for women and men to contribute both in volunteer and working positions. The project has HR policy and same is followed equally. The project owner has a specified HR policy that considers participation by both men and women.* *Besides, China also ratified relevant ILO core conventions on equality, namely Equal Remuneration Convention (Convention No 100).* *Therefore, it is confirmed equal opportunities have been provided by the project activity in the context of gender equity and pay for labor and work.*

Forced labor	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Criminal Law", which could prohibit such phenomena.* *Forced labor is an illegal activity in the host country and the local labor compliance takes into account of the same. Further, China is a party to ILO and forced labour is illegal in China.* *China has relevant laws and regulations in place prohibiting forced and compulsory labor, such as the Labour Contract Law, Women's Rights Protection Law, Prohibition of Child Labor Regulation. Via checking HR Records and interviewing staff from the project owner, it was verified the project fully complies with the relevant laws and respects fundamental right of employee.* *China has strict prohibition for child labour. The project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview.* *By site visit, it is confirmed no human trafficking, forced labor, and child labor detected.*
Child labor	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Criminal Law", which could prohibit such phenomena.* *Forced labor is an illegal activity in the host country and the local labor compliance takes into account of the same. Further, China is a party to ILO and forced labour is illegal in China.* *China has relevant laws and regulations in place prohibiting forced and compulsory labor, such as the Labour Contract Law, Women's Rights Protection Law, Prohibition of Child Labor Regulation. Via checking HR Records and interviewing staff from the project owner, it was verified the project fully complies with the relevant laws and respects fundamental right of employee.* *China has strict prohibition for child labour. The project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was*

	confirmed by the assessment team through checking HR Records and Staff Roaster and site interview. *By site visit, it is confirmed no human trafficking, forced labor, and child labor detected.*
Human trafficking	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights and equity. China has published "Labor Law" and "Criminal Law", which could prohibit such phenomena.* *Forced labor is an illegal activity in the host country and the local labor compliance takes into account of the same. Further, China is a party to ILO and forced labour is illegal in China.* *China has relevant laws and regulations in place prohibiting forced and compulsory labor, such as the Labour Contract Law, Women's Rights Protection Law, Prohibition of Child Labor Regulation. Via checking HR Records and interviewing staff from the project owner, it was verified the project fully complies with the relevant laws and respects fundamental right of employee.* *China has strict prohibition for child labour. The project does not involve child labour neither in construction nor at operation of project activity. In the HR policy is also mentioned that no child labour will be entertained. The same was confirmed by the assessment team through checking HR Records and Staff Roaster and site interview.* *By site visit, it is confirmed no human trafficking, forced labor, and child labor detected.*

3.2.3.2 Human Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
None	*By checking Project approval and EIA approval issued by government, it is confirmed the project has been approved by Chinese government, in this sense, the project should meet all the requirement of regulation and laws set up by Chinese government in aspect of human rights* *China has its own legislation in place prohibiting the violation of human rights principle and it actively enforces the compliance of such principle. China has ratified two UN human rights treaties--the International Covenant on Civil and*

	Political Rights, the International Covenant on Economic, Social, and Cultural Rights. China has published relevant regulations and practice into line with the treaties, such as Labor Law. *Therefore, it is confirmed that the project does not discriminate with regards to participation and inclusion. The project abides the rules of equality accordingly and does not involve and is not complicit in any form of discrimination. Specific to gender equality, detailed enforcement rules are regulated in "Women's Rights Protection Law" to provide support and benefits to women.*

3.2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
None	*By checking EIA report approved by government, it is confirmed that no indigenous peoples present in or within the area of influence of the project, the project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.* *Hence not applicable for the project.*

3.2.3.4 Property Rights

Risks identified	Evidence gathering activities, evidence checked, and assessment conclusion
None	*By checking EIA report approved by government, it is confirmed that no indigenous peoples present in or within the area of influence of the project, the project area does not include sites, structures, or objects with historical, cultural, artistic, traditional or religious values or intangible forms of culture.* *Hence not applicable for the project.*

3.2.3.5 Benefit Sharing

Item	Evidence gathering activities, evidence checked, and assessment conclusion

Process used to design the benefit sharing plan	*Not applicable as no property rights involved for the project.*
Summary of the benefit sharing plan	*Not applicable as no property rights involved for the project.*
Approval and dissemination of benefit sharing plan	*Not applicable as no property rights involved for the project.*

3.2.4 Ecosystem Health

Item	Evidence gathering activities, evidence checked, and assessment conclusion
Impacts on biodiversity and ecosystems	*By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.*
Soil degradation and soil erosion	*The Project's purpose is to supply electricity from the landfill gas. Therefore, the impact of the Project on soil degradation and soil erosion is negligible.* *By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.*
Water consumption and stress	*The Project's purpose is to supply electricity from the landfill gas. Therefore, the impact of the Project on water consumption and stress is negligible.* *By checking EIA report approved by government, it is confirmed that the project has no negative impacts on biodiversity and ecosystems.*

3.2.4.1 Rare, Threatened, and Endangered species

Risk identified	Evidence gathering activities, evidence checked, and assessment conclusion
Species and habitat	*By checking EIA report approved by government, it is confirmed that the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.*

Areas needed for habitat connectivity	*By checking EIA report approved by government, it is confirmed that the project isn't located in, or adjacent to habitats for rare, threatened, or endangered species.*

3.2.4.2 Introduction of Species

Species introduced	Evidence gathering activities, evidence checked, and assessment conclusion
Not applicable	*Not applicable*

Existing invasive species	Evidence gathering activities, evidence checked, and assessment conclusion
Not applicable	*Not applicable*

	Evidence gathering activities, evidence checked, and assessment conclusion
Invasive species	*Not applicable*

3.2.4.3 Ecosystem conversion

Risks Identified	Evidence gathering activities and evidence checked
Ecosystem conversion	*Not applicable as the project is not a ARR, ALM, WRC or ACoGS project.*

3.3 Application of Methodology

3.3.1 Title and Reference

Assessment team checked that following methodology and tools are applicable for the project activity. The details are as below:

Title: *Landfill methane recovery; Grid connected renewable electricity generation*

Reference: *The project activity meets the eligibility criteria of small-scale project as the estimated annual average GHG emission reductions or removal per year is 35,258 tCO_2e which is less than 60,000 tonnes of CO_2e per year.*

Methodology: *Landfill methane recovery, AMS-III.G, version 10.0*

Grid connected renewable electricity generation, AMS-I.D, version 18.0

Type I and III: *Energy industries (renewable / non-renewable sources); Waste handling and disposal*

Sectoral scope(s): *01 and 13*

Tools referred with above methodology and applicable for project activity are:

- Tool 05: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation, version 03.0

- Tool 04: Emissions from solid waste disposal sites, version 08.1

- Tool 07: Tool to calculate the emission factor for an electricity system, version 07.0

- Tool 32: Positive lists of technologies, version 04.0

3.3.2 Applicability

Methodology ID	Applicability condition	Assessment and conclusion
AMS-III.G	*2. This methodology comprises measures to capture and combust methane from landfills (i.e. solid waste disposal sites) used for the disposal of residues from human activities including municipal, industrial, and other solid wastes containing biodegradable organic matter.*	*Applicable.* *By checking the FSR /15/, Project approval and on-site audit, it is confirmed that the Project consists in capturing landfill gas (which contains methane) from the Suide Landfill site, which is used for disposal of residues from human activities.*
AMS-III.G	*3. Different options to utilise the recovered landfill gas as detailed in paragraph 4 for "AMS-III.H. : Methane recovery in wastewater treatment" (version 19.0) are eligible for use under this methodology. The relevant procedures in AMS-III.H. shall be followed in this regard.* *Paragraph 4 of AMS-III.H.:*	*Applicable.* *By checking FSR and site visit interview, it is confirmed that the project utilizes the recovered LFG to generate electrical energy directly which meet the requirement of 4 (a) of AMS-III.H. (version 19.0).*

	The recovery biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: *(a) Thermal or mechanical, electrical energy generation directly;* *(b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or* *(c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed:* *(i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints;* *(ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users;* *or(iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users* *(d) Hydrogen production;*	

	(e) Use as fuel in transportation applications after upgrading.	
AMS-III.G	*4. Measures are limited to those that result in aggregate emission reductions of less than or equal to 60 kt CO_2 equivalent annually from all Type III components of the project activity.*	*Applicable.* *It is confirmed the annual ER of the project is less than 60,000 CO_2e from all type III components.*
AMS-III.G	*5. The proposed project activity does not reduce the amount of organic waste that would have been recycled in the absence of the project activity*	*Applicable.* *By checking FSR and site visit interview, it is confirmed that the implementation of the project does not reduce the amount of organic waste that would be recycled in the absence of the project. All the solid waste is disposed in the Suide Landfill site.*
AMS-III.G	*6. This methodology is not applicable if the management of the solid waste disposal site SWDS) in the project activity is deliberately changed in order to increase methane generation compared to the situation prior to the implementation of the project activity (e.g. other than to meet a technical or regulatory requirement). Such changes may include, for example, the addition of liquids to a SWDS, pre-treating waste to seed it with bacteria for the purpose of increasing the rate of anaerobic degradation of the*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed that this condition is not applicable for the project.*

	SWDS or changing the shape of the SWDS to increase methane production	

Methodology ID	Applicability condition	Assessment and conclusion
AMS-I.D	*4. This methodology is applicable to project activities that:* *(a) Install a Greenfield plant;* *(b) Involve a capacity addition in (an) existing plant (s);* *(c) Involve a retrofit of (an) existing plant (s);* *(d) Involve a rehabilitation of (an) existing plant (s)/unit (s) or;* *Involve a replacement of (an) existing plant(s).*	*Applicable.* *By checking FSR and site visit interview, it is confirmed that the project is a greenfield project.*
AMS-I.D	*5. Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:* *(a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir;* *(b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project*	*Applicable.* *By checking FSR and site visit interview, it is confirmed this is not appliable as project is not a hydro power project.*

	activity, as per definitions given in the project emissions section, is greater than 4W/m²; *The project activity results in new reservoirs and the power density of the powerplant, as per definitions given in the project emissions section, is greater than 4W/m².*	
AMS-I.D	*6. If the new unit has both renewable and non-renewable components (e.g. a wind/ diesel unit), the eligibility limit of 15 MW for a small-scale CDM project activity applies only to the renewable component. If the new unit co-fires fossil fuel. the capacity of the entire unit shall not exceed the limit of 15 MW.*	*Applicable.* *By checking FSR and site visit interview, it is confirmed the total installed capacity of the project is 1 MW.*
AMS-I.D	*7. Combined heat and power (co-generation) systems are not eligible under this category.*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed the project does not involve in heat generation.*
AMS-I.D	*8. In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed the project does not involve in capacity addition.*

AMS-I.D	*9. In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed the project does not involve in retrofit, rehabilitation or replacement.*
AMS-I.D	*10. In the case of landfill gas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-L.C. Thermal energy production with or without electricity shall be explored.*	*Applicable.* *By checking FSR and site visit interview, it is confirmed the project is designed to install four engines to combusted LFG of Suide Landfill, which mainly contains 50% methane.*
AMS-I.D	*11. In case biomass Is sourced from dedicated plantations, the applicability criteria in the tool "Project emissions from cultivation of biomass "shall apply.*	*Not applicable.* *By checking FSR and site visit interview, it is confirmed that the project does not involved in biomass.*

Methodology ID	Applicability condition	Assessment and conclusion

Tool 04 Emissions from solid waste disposal sites	*(a) Application A: The VCS project activity mitigates methane emissions from a specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "AMS-III.G: Landfill methane recovery). The methane is generated from waste disposed in the past, including prior to the start of the VCS project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (VCS-PD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e. g. measuring the amount of methane captured from the SWDS).*	*Applicable. The project collects the methane from the Suide landfill site and uses it to generate the electricity.*
Tool 04 Emissions from solid waste disposal sites	*(b) Application B: The VCS project activity avoids or involves the disposal of waste at SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW) is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided*	*Not applicable. The project collects the methane from the Suide landfill site and uses it to generate the electricity.*

	from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions.	

Methodology ID	Applicability condition	Assessment and conclusion
Tool 07 Tool to calculate the emission factor for an electricity system	*This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).*	*Applicable, this tool is applied to estimate the OM, BM and/or CM.*
Tool 07 Tool to calculate the emission factor for an electricity system	*Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the Project proponents, i.e. option IIa and option IIb. If option IIa is chosen, the conditions*	*Applicable, the emission factor for the project electricity system is calculated for grid power plants only.*

	specified in "Appendix 1: Procedures related to off-grid power generation" should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
Tool 07 Tool to calculate the emission factor for an electricity system	*In case of VCS projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.*	*Not applicable, the project electricity system is located in China.*
Tool 07 Tool to calculate the emission factor for an electricity system	*Under this tool, the value applied to the CO_2 emission factor of biofuels is zero.*	*In case there are biofuel involved, the value applied to the CO_2 emission factor of biofuels is zero.*

Methodology ID	Applicability condition	Assessment and conclusion
Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream	*Typical applications of this tool are methodologies where the flow and composition of residual or flared gases or exhaust gases are measured*	*Applicable. The project collects the methane from the Suide landfill site and uses it to generate the electricity. The flow of*

	for the determination of baseline or project emissions.	*methane is measured for the determination of baseline emissions.*
Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream	*Methodologies where CO_2 is the particular and only gas of interest should continue to adopt material balances as the means of flow determination and may not adopt this tool as material balances are the cost effective way of monitoring flow of CO_2.*	*Not applicable.*
Tool 08 Tool to determine the mass flow of a greenhouse gas in a gaseous stream	*The underlying methodology should specify:* *(a) The gaseous stream the tool should be applied to;* *(b) For which greenhouse gases the mass flow should be determined;* *(c) In which time intervals the flow of the gaseous stream should be measured; and* *(d) Situations where the simplification offered for calculating the molecular mass of the gaseous stream (equations (3) or (17)) is not valid (such as the gaseous stream is predominantly composed of a gas other than N_2).*	*Applicable.* *It is confirmed that the gaseous stream determined the baseline emissions is LFG which mainly contains fraction of CH_4, which corresponds to point (a) and (b).* *LFG combusted by the engines is measured continuously by a flow meter and recorded hourly, which corresponds to point (c).* *Point (d) is not applicable.*

Methodology ID	Applicability condition	Assessment and conclusion
Tool 32 Positive lists of technologies	*The use of this methodological tool is not*	*Applicable.*

	mandatory for the Project proponents of a VCS project activity for demonstrating their additionality.	The project apply tool for demonstrating its additionality
Tool 32 Positive lists of technologies	This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Applicable. The project apply tool conjunction with large-scale methodology AMS-III.G which refers to this tool.
Tool 32 Positive lists of technologies	The positive lists as contained in section 5 of this tool are valid up to 10/03/2025. Notwithstanding the provisions on the validity of new, revised and previous versions of methodologies and methodological tools in the "Procedure: Development, revision and clarification of baseline and monitoring methodologies and methodological tools", there will be no grace period for the application of this tool and the validity of the positive list after this date, including in cases where further technologies are added to the positive list through revisions of this tool before this date.	Applicable. The project collects the LFG from the Suide landfill site and uses it to generate the electricity, with a total capacity of 1 MW, which conforms to 5.1.1 (a) "The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW".

3.3.3 Project Boundary

As per AMS-III.G and AMS-I.D, the project boundary is the physical, geographical site of the landfill where the gas is captured and destroyed/used and the project power plant and all power plants connected physically to the electricity system that the project power plant is connected to.

It is confirmed by the assessment team that the project boundary of the project activity includes the sites where the LFG is flared or used (e.g. power plant). The electricity used by the project *activity is sourced from NWCPG and electricity generated by LFG capturing of the project activity is connected to NWCPG. Therefore, the project boundary includes the whole LFG related system (e.g. LFG collection, LFG pre-treatment system, LFG power generation system, etc.) and all grid-connected power plants in NWCPG. According to China DNA, NWCPG covers Shaanxi Province, Gansu Province, Qinghai Province, Ningxia Autonomous Region, Xinjiang Autonomous Region.*

The sources and GHG gases involved for the Project activity are as below:

Source		Gas	Included?	Justification/Explanation
Baseline	*Emissions from decomposition of waste at the SWDS site*	CO_2	*No*	*CO_2 emissions from decomposition of organic waste are not accounted since the CO_2 is also released under the project activity*
		CH_4	*Yes*	*The major source of emissions in the baseline*
		N_2O	*No*	*N_2O emissions are small compared to CH_4 emissions from SWDS. This is conservative*
	Emissions from electricity generation	CO_2	*Yes*	*Major emission source*
		CH_4	*No*	*Excluded for simplification. This is conservative*
		N_2O	*No*	*Excluded for simplification. This is conservative*
	Emissions from heat generation	CO_2	*No*	*No heat generation is included in the project activity*
		CH_4	*No*	*No heat generation is included in the project activity*
		N_2O	*No*	*No heat generation is included in the project activity*
	Emissions from the use of natural gas	CO_2	*No*	*No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity*
		CH_4	*No*	*No supply of LFG through a natural gas distribution network, dedicated pipeline or using trucks is included in the project activity*
		N_2O	*No*	*No supply of LFG through a natural gas distribution network, dedicated pipeline or*

Source		Gas	Included?	Justification/Explanation
				using trucks is included in the project activity
Project	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO_2	No	Excluded for no fossil fuel is used by the project activity
		CH_4	No	Excluded for no fossil fuel is used by the project activity
		N_2O	No	Excluded for no fossil fuel is used by the project activity
	Emissions from electricity consumption due to the project activity	CO_2	Yes	May be an important emission source
		CH_4	No	Excluded for simplification. This emission source is assumed to be very small
		N_2O	No	Excluded for simplification. This emission source is assumed to be very small
	Emissions from flaring	CO_2	No	Excluded for simplification
		CH_4	No	Excluded for simplification
		N_2O	No	Excluded for simplification
	Emissions from distribution of LFG using trucks and dedicated pipelines	CO_2	No	No distribution of LFG using trucks and dedicated pipelines
		CH_4	No	No distribution of LFG using trucks and dedicated pipelines
		N_2O	No	No distribution of LFG using trucks and dedicated pipelines

3.3.4 Baseline Scenario

As per AMS-III.G, the baseline scenario for LFG is the situation where, in the absence of the project activity, biomass and other organic matter are left to decay within the project boundary, and methane is emitted to the atmosphere, possibly with capture of LFG and destruction through flaring to comply with regulations or contractual requirements. For electricity, if all or part of the electricity generated by the project activity is exported to the grid, the baseline scenario for all or the part of the electricity exported to the grid is assumed to be electricity generation in existing and/or new grid-connected power plants.

The project activity captures LFG to produce electricity and supplies to the grid. In the absence of the project, LFG from Suide landfill is emitted directly into atmosphere, the equivalent amount of power would have been supplied by CCPG, which is fed mainly by fossil fuel fired plants.

Applus+ Certification confirms that the Project has been approved by Chinese government by checking the Project approval and Environmental Impact Assessment (EIA) approval. By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.

The assessment team, therefore, concludes that the VCS PD conforms to the guidance given by EB via CDM Validation and Verification Standard for project activities version 03.0 and VCS via VCS Standard version 4.7.

The project activity captures LFG to produce electricity and supplies to the grid. According to GB50869-2013 has been implemented from 01/03/2014, Item 3.0.3, 4.0.2, 8.1.1, 10.1.1, 11.1.1, 11.6.1, 11.6.3, 11.6.4 and 15.0.5 are mandatory provisions and must be strictly implemented. But Item 11.1.3 is a voluntary provision, and it is a common practice in China that the LFG from landfill sites is vented to the atmosphere directly. And the assessment team confirmed that no regulatory surplus required for project activity.

The assessment team confirmed the baseline of the project is in the absence of the project, LFG from Suide landfill site is emitted directly into atmosphere, the equivalent amount of power would have been supplied by NWCPG, which is fed mainly by fossil fuel fired plants.

Applus+ Certification confirms that the Project has been approved by Chinese government by checking the Project approval and Environmental Impact Assessment (EIA) approval. By checking laws and regulation, it is confirmed that the project activity is in complicate with all laws and regulations in China.

3.3.5 Additionality

Assessment for regulatory surplus

Based on the professional knowledge of the assessment team, it is able to confirm there are no regulatory surplus for LFG generation project in China.

Assessment of Prior consideration of carbon revenue

Following time lime of project has been assessed and confirmed to be accurate.

Events	***Time***	***Assessment***
Completion of FSR	*12/2021*	*Confirmed by checking FSR*
FSR approval	*14/01/2022*	*Confirmed by checking Project Approval*
Board resolution	*24/01/2022*	*Confirmed by checking Board resolution /02/*

Completion of Environmental Impact Assessment Report	*07/2022*	*Confirmed by checking EIA report*
EIA approval	*24/08/2022*	*Confirmed by checking EIA Approval*
VCS stakeholder consultation meeting	*13/09/2022*	*Confirmed by checking Stakeholder Meeting Minutes*
Construction date of the project	*30/09/2022*	*Confirmed by checking Construction Order*
The project was put into operation.	*24/02/2023*	*Confirmed by checking Operation log and site visit*

Additionality Assessment

As per the Methodological Tool "Positive lists of technologies", the project activities at new or existing landfills (greenfield or brownfield) are deemed automatically additional, if it is demonstrated that prior to the implementation of the project activities the landfill gas (LFG) was only vented and/or flared (in the case of brownfield projects) or would have been only vented and/or flared (in the case of greenfield projects) but not utilized for energy generation, and that under the project activities any of the following conditions are met:

(a) The LFG is used to generate electricity in one or several power plants with a total nameplate capacity that equals or is below 10 MW;

(b) The LFG is used to generate heat for internal or external consumption;

(c) The LFG is flared.

The project activity uses LFG which is collected from Suide landfill site to generate electricity. Prior to the implementation of the project activity, the LFG from Suide landfill site was only vented, not utilized for energy generation. By interviewing local stakeholders, staff from local government, staff from Suide landfill site and checking website of local government, it is confirmed by the assessment team that only the project activity uses LFG from Suide landfill site to generate electricity. By checking Project Approval and EIA Approval, the total installed capacity of the project activity is 1 MW, which is below 10 MW. Therefore, the project activity is deemed automatically additional.

3.3.6 Quantification of GHG Emission Reductions and Carbon Dioxide Removals

Assessment team checked the baseline, project and leakage calculation and confirm that the evaluation of baseline, project and leakage is as per the approved methodology and formula used to calculate the same is correct. The detail analysis is as below:

Baseline emissions

Baseline emissions are determined according to the following equation for LFG part:

$$BE_{y,1}=\eta_{PJ}\times BE_{CH4,SWDS,y}-(1-OX)\times F_{CH4,BL,y}\times GWP_{CH4}$$

Where:

$BE_{y,1}$	=	*Baseline emissions associated with the SWDS in year y (tCO_2e/yr)*
η_{PJ}	=	*Efficiency of the LFG capture system that will be installed in the project activity. It is used for ex ante estimation only. A default value of 50 per cent may be used*
$BE_{CH4,SWDS,y}$	=	*Methane emission potential of a solid waste disposal site (tCO_2)*
OX	=	*Oxidation factor (reflecting the amount of methane from SWDS that is oxidised in the soil or other material covering the waste) (dimensionless). A default value of 0.1 may be used*
$F_{CH4,BL,y}$	=	*Methane emissions that would be captured and destroyed to comply with national or local safety requirement or legal regulations in the year y (tCH_4)*
GWP_{CH4}	=	*Global Warming Potential for methane*

For the project, Methane emission potential of a solid waste disposal site ($BE_{CH4,SWDS,y}$) is calculated using first order decay (FOD) model as follows:

$$BE_{CH4,SWDS,y}=\varphi_y\times(1-f_{y)}\times GWP_{CH4}\times(1-OX)\times\frac{16}{12}\times F\times DOC_{f,y}\times MCF_y\times\sum_{x=1}^{y}\sum_{j}(W_{j,x}\times DOC_j\times e^{-k_j\times(y-x)}\times(1-e^{k_j})$$

Where:

$BE_{CH4,SWDS,y}$	=	*Baseline, project or leakage methane emissions occurring in year y generated from waste disposal at a SWDS during a time period ending in year y (tCO_2e/yr)*
x	=	*Years in the time period in which waste is disposed at the SWDS, extending from the first year in the time period (x = 1) to year y (x = y)*
y	=	*Year of the crediting period for which methane emissions are calculated (y is a consecutive period of 12 months)*
$DOC_{f,y}$	=	*Fraction of degradable organic carbon (DOC) that can decomposes under the specific conditions occurring in the SWDS for year y (weight fraction)*

$W_{j,x}$ = Amount of solid waste type j disposed or prevented from disposal in the SWDS in the year x (t)

φ_y = Model correction factor to account for model uncertainties for year y

f_y = Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y

GWP_{CH4} = Global Warming Potential of methane

OX = Oxidation factor (reflecting the amount of methane from SWDS that is oxidized in the soil or other material covering the waste)

F = Fraction of methane in the SWDS gas (volume fraction)

MCF_y = Methane correction factor

DOC_j = Fraction of degradable organic carbon in the waste type j (weight fraction)

k_j = Decay rate for the waste type j (1 / yr)

j = Type of residual waste or types of waste in the MSW

The parameters required to apply the FOD model is determined as:

Parameter	Application A	Justification
ϕ_y	0.75	Baseline emissions: default values
OX	0.1	Default value
F	0.5	Default value
$DOC_{f,y}$	0.5	Default value
MCF_y	1.0	Default values (based on SWDS type)
k_j	Refer to Section 3.3.8	Default values (based on waste type)
$W_{j,x}$	Refer to Section 3.3.8	Estimated once, with the waste composition obtained from EIA.
DOC_j	Refer to Section 3.3.8	Default values (based on waste type)
f_y	0	Requirement from AMS-III.G and Tool 04

Determination of $F_{CH4,BL,y}$

This section provides a procedure to determine the amount of methane that would have been captured and destroyed (by flaring) in the baseline due to regulatory or contractual requirements, to address safety and odour concerns, or for other reasons (collectively referred to as *requirement in this section). The four cases in the following table are distinguished. The appropriate case should be identified, and the corresponding instructions followed.*

Situation at the start of the project activity	*Requirement to destroy methane*	*Existing LFG capture and destruction system*
Case 1	*No*	*No*
Case 2	*Yes*	*No*
Case 3	*No*	*Yes*
Case 4	*Yes*	*Yes*

Currently China has regulations in place to deal with the management of landfills and to encourage utilization of LFG. Those regulations are:

“Standard for Pollution Control on the Landfill Site of Municipal Solid Waste” (GB 16889-2008), which became effective in 2008, issued by the Environment Protection Administration.

“Technical Code for Municipal Solid Waste Sanitary Landfill” (GB 50869-2013), issued by the Ministry of Construction in 2013.

According to item 5.15 of GB16889-2008, if the designed landfill capacity is more than 2.5 million tons and the landfill thickness is more than 20m, methane utilization facilities or flare burning facilities shall be built to treat the landfill gas containing methane. For municipal solid waste landfills smaller than the above scale, technologies that can effectively reduce methane generation and emission shall be adopted or flare combustion facilities shall be used to treat methane containing landfill gas.

Item 11.1.1 of GB 50869-2013 stipulates that the landfill site must be equipped with effective landfill gas drainage facilities to prevent the natural accumulation and migration of landfill gas, causing fire and explosion. Item 11.1.3 stipulates that if the landfill does not have the conditions for landfill gas utilization, the flare method shall be adopted for combustion treatment, and the process that can effectively reduce the generation and emission of methane shall be adopted. The old landfills that are not safe and stable should be equipped with effective landfill gas drainage facilities. Among them, item 11.1.1 is mandatory and must be strictly implemented.

In fact, the LFG of Suide Landfill is emitted to atmosphere without LFG capture system prior the implementation of the project. Therefore, Case 2 listed in the table above is applicable for the project.

The requirements above don’t specify any amount or percentage of LFG that should be destroyed. In this situation:

$F_{CH4,BL,y}=F_{CH4,BL,R,y}$

$F_{CH4,BL,R,y}=\rho_{reg,y}\times F_{CH4,PJ,capt,y}$

Where:

$F_{CH4,BL,R,y}$	=	*Amount of methane in the LFG which is flared in the baseline due to a requirement in year y (tCH_4/yr)*
$\rho_{reg,y}$	=	*Fraction of LFG that is required to be flared due to a requirement in year y. A default value of 20% may be used.*
$F_{CH4,PJ,capt,y}$	=	*Amount of methane in the LFG which is captured in the project activity in year y (tCH_4/yr)*

<u>Baseline emissions are determined according to the following equation for electricity part:</u>

$BE_{y,2}=EG_{PJ,y}\times EF_{grid,y}$

$BE_{y,2}$	=	*Baseline emissions associated with electricity generation in year y (tCO_2)*
$EG_{PJ,y}$	=	*Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the CDM project activity in year y (MWh)*
$EF_{Grid,y}$	=	*Combined margin CO_2 emission factor for grid connected power generation in year y calculated using the latest version of the "Tool to calculate the emission factor for an electricity system" (tCO_2/MWh)*

Calculation of $EG_{PJ,y}$

The project is the installation of a greenfield power plant, therefore:

$EG_{PJ,y}=EG_{PJ,facility,y}$

Where:

$EG_{PJ,facility,y}$	=	*Quantity of net electricity generation supplied by the project plant/unit to the grid in year y (MWh)*

<u>Determination of the emission factor for electricity generation ($EF_{Grid,y}$)</u>

The baseline scenario of the project is that the LFG from Suide MSW landfill site was released directly into atmosphere and equivalent electricity generation by the project was supplied by CCPG, Scenario A of the "Methodological tool: Baseline, project and/or leakage emissions from electricity consumption and monitoring of electricity generation" applies. $EF_{Grid,y}$ shall therefore be determined in accordance with Option A1 of the tool, i.e. the applied emission factor shall be the combined margin emission factor of the CCPG, calculated in accordance with the "Tool to calculate the emission factor of an electricity system" ($EF_{Grid,y}$ = $EF_{grid,CM,y}$).

The grid emission factor is calculated as the weighted average of the operating margin (0.5) & build margin (0.5) values. The value of combined margin is sourced from 2023 Baseline Emission Factors for Regional Power Grids in China dated 09/07/2024 published by China DNA. China DNA calculates the data based on Tool to Calculate the Emission Factor for an Electricity System", version 07.0. No further assessment is required for grid emission calculation as the ex-ante value is sourced directly from the Chinese DNA.

Emission factor ($EF_{Grid,y}$): $EF_{Grid,y} = EF_{grid,CM,y} = 0.9014 \times 0.5 + 0.3597 \times 0.5 = 0.63055\ tCO_2/MWh$.

This value is fixed ex-ante for the crediting period.

ID number	$F_{CH4,PJ,y}$	$F_{CH4,BL,y}$	$BE_{y,1}$
Unit	***tCH_4***	***tCH_4***	***tCO_2e***
24/02/2023 – 31/12/2023	*901*	*180*	*20,691*
2024	*1,141*	*228*	*26,194*
2025	*1,223*	*245*	*28,072*
2026	*1,303*	*261*	*29,919*
2027	*1,383*	*277*	*31,747*
2028	*1,461*	*292*	*33,548*
2029	*1,539*	*308*	*35,332*
2030	*1,616*	*323*	*37,101*
2031	*1,536*	*307*	*35,269*
2032	*1,460*	*292*	*33,531*
01/01/2033 – 23/02/2033	*205*	*41*	*4,717*

ID number	$EG_{PJ,y}$	$EF_{grid,CM,y}$	$BE_{y,2}$
Unit	*MWh*	***tCO_2e/MWh***	***tCO_2e***
24/02/2023 – 31/12/2023	*3,786*	*0.63055*	*2,387*
2024	*4,793*	*0.63055*	*3,022*
2025	*5,137*	*0.63055*	*3,238*

2026	5,475	0.63055	3,451
2027	5,809	0.63055	3,662
2028	6,139	0.63055	3,870
2029	6,465	0.63055	4,076
2030	6,789	0.63055	4,280
2031	6,453	0.63055	4,069
2032	6,135	0.63055	3,868
01/01/2033 – 23/02/2033	863	0.63055	544

Year	Emission Reduction by Methane recovery from the landfill in the absence of the project activity during the crediting period (tCO_2e) ($BE_{y,1}$)	Emission Reduction from Electricity Generation (tCO_2e) ($BE_{y,2}$)	Baseline Emissions (tCO_2e)
24/02/2023 – 31/12/2023	20,691	2,387	23,078
2024	26,194	3,022	29,216
2025	28,072	3,238	31,310
2026	29,919	3,451	33,370
2027	31,747	3,662	35,409
2028	33,548	3,870	37,418
2029	35,332	4,076	39,408
2030	37,101	4,280	41,381
2031	35,269	4,069	39,338
2032	33,531	3,868	37,399
01/01/2033 – 23/02/2033	4,717	544	5,261

Total (tones of CO_2e)	*316,121*	*36,467*	*352,588*

Project emissions

As per AMS-III.G, Project emissions are calculated as follows:

$$PE_y=PE_{power,y}+PE_{flare,y}+PE_{process,y}$$

Where:

PE_y	=	*Project emissions in year y (tCO_2/yr)*
$PE_{power,y}$	=	*Emissions from the use of fossil fuel or electricity for the operation of the installed facilities in the year y (tCO_2e)*
$PE_{flare,y}$	=	*Emissions from flaring or combustion of the landfill gas stream in the year y (tCO_2e)*
$PE_{process,y}$	=	*Emissions from the landfill gas upgrading process in the year y (tCO_2e), determined by following the relevant procedures described in annex 1 of AMS-III.H.*

According to AMS-III.G, project emissions from electricity consumption are determined as per the procedures described in AMS-I.D "Grid connected renewable electricity generation". According to AMS-I.D, the project is neither involved geothermal power plants nor hydro power plants, project emissions from electricity consumption are identified as 0. Besides, as fossil fuel is not be used by the project, project emissions from fossil fuel consumption are also 0. Therefore, $PE_{power,y}$ *is equal to 0 for ex-ante estimation.*

There is no flare used to destroy the LFG, therefore, $PE_{flare,y}$ *Is equal to 0.*

The project captures LFG for power generation and not involved upgrading process, therefore, $PE_{process,y}$ *is equal to 0.*

Therefore, $PE_y = 0\ tCO_2e$

Leakage

As the methane recovery technology is not equipment transferred from another activity, leakage effects are to be considered. The validation team deems this consideration is correct and in line with methodology AMS-III.G, version 10.0 and AMS-I.D, version 18.0.

Therefore, $LE_y = 0\ tCO_2e$

Emission Reductions

According to AMS-III.G, version 10.0 and AMS-I.D, version 18.0, emission reductions are calculated as follows:

$ER_y = BE_y - PE_y - LE_y$

Where:

ER_y	=	*Emission reductions in year y (tCO_2e/yr)*
BE_y	=	*Baseline emissions in year y (tCO_2e/yr)*
PE_y	=	*Project emissions in year y (tCO_2e/yr)*
LE_y	=	*Leakage in year y (tCO_2e/yr)*

Hence for the project activity, the estimated amount of GHG emission reductions (ER_y) is 247,728 tCO_2e during the crediting period from 01/01/2022 to 31/12/2031, resulting in estimated average annual emission reductions of 24,772 tCO_2e.

Year	Estimated baseline emissions or removals (tCO_2e)	Estimated project emissions or removals (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated net GHG emission reductions or removals (tCO_2e)
24/02/2023 – 31/12/2023	23,078	0		23,078
2024	29,216	0		29,216
2025	31,310	0		31,310
2026	33,370	0		33,370
2027	35,409	0		35,409
2028	37,418	0		37,418
2029	39,408	0		39,408
2030	41,381	0		41,381
2031	39,338			39,338
2032	37,399			37,399
01/01/2033 – 23/02/2033	5,261			5,261

	352,588			352,588

3.3.7 Methodology Deviations

No methodology deviation is applied in the project.

Total	*0*	*0*

3.3.8 Monitoring Plan

Parameters determined ex-ante

Data /	
Data u	
Description:	*Oxidation factor (reflecting the amount of methane from ...dized in the soil or other material covering the*
Source of data used:	*Consistent with an extensive review of published literature on this subject, including the 2019 Refinement to the 2006 IPCC ...ational Greenhouse Gas Inventories*
Value applied:	*0.1*

Data / Parameter:	*GWP_{CH4}*
Data unit:	*$tCO_2e/t\ CH_4$*
Description:	*Global warming potential of CH_4*
Source of data used:	*IPCC Fifth Assessment Report (AR5)*
Value applied:	*28*

Data / Parameter:	*D_{CH4}*
Data unit:	*t/m^3*
Description:	*Density of methane gas at Normal Conditions*
Source of data used:	*Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0*
Value applied:	*0.000716*

Data / Parameter:	*η_{PJ}*

Data u	
Description:	*Efficiency of the LFG capture system that will be installed in the project activity*
Source of data used:	*FSR of the project*
Value a	

Data /	
Data u	
Descri	*rrection factor to account for model*
Source	*the tool "Emissions from solid waste disposal 8.1)*
Value applied:	*0.75*

Data / Parameter:	*f_y*
Data unit:	*Dimensionless*
Description:	*Fraction of methane captured at the SWDS and flared, combusted or used in another manner that prevents the emissions of methane to the atmosphere in year y*
Source of data used:	*As in the baseline scenario, there is no LFG capturing and usage, zero is considered and in line with the requirement of tool "Emissions from solid waste disposal sites" (version 08.1)*
Value applied:	*0*

Data / Parameter:	*F*
Data unit:	*/*
Description:	*Fraction of methane in the SWDS gas (volume fraction)*
Source of data used:	*2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories*
Value applied:	*0.5*

Data /

Data u

Descri r the fraction of degradable organic carbon at decomposes in the SWDS

Source of data used: 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

Value a

Data /

Data u

Descri ion factor

Source of data used: 2019 Refinement to the 2006 IPCC Guidelines for National Inventories

Value applied: 1.0

Suide MSW landfill site is an anaerobic managed solid waste disposal site.

As confirmed through site visit and checking FSR of the project, Suide MSW landfill site has controlled placement of waste by mechanical compacting and levelling of the waste, then value 1.0 has been applied

Data / Parameter: DOC_j

Data unit: /

Description: Fraction of degradable organic carbon in the waste type j (weight fraction)

Source of data used: 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories

Value applied: For MSW, the following values for the different waste types j should be applied:

Waste type j	DOC_j (% wet waste)
Wood and wood products	43
Pulp, paper and cardboard (other than sludge)	40

Textiles	24
Garden, yard and park waste	20
Glass, plastic, metal, other inert waste	0

Data / Parameter:	k_j
Data unit:	
Description:	e waste type j
Source of data:	nt to the 2006 IPCC Guidelines for National Inventories
Value applied:	Apply the following default values for the different waste types j:

Waste type j		k_j (MAT ≤20 ℃, MAP/PET<1)
Slowly degrading	Pulp, paper, cardboard (other than sludge)	0.04
	Wood, wood products and straw	0.02
Moderately degrading	Other (non-food) organic putrescible garden and park waste	0.05
Rapidly degrading	Food, food waste, sewage sludge, beverages and tabacco	0.06

As per the FSR of the project activity and online information, the climate data for the location of Suide MSW landfill is:

Temperature: 10.4 ℃

MAP – Mean annual precipitation: 452.5 mm

PET – Potential evapotranspiration: 2,069 mm

Data / Parameter:	$W_{j,x}$
Data unit:	t

Descri… …nic waste type j disposed/prevented from …WDS in the year y

Source of data used: Estimated once in the FSR

…nt of waste is sourced from FSR based on … forecast for the future situation. Considering …iled by third-party with accreditation and …overnment as project approval. FSR is … reliable source for data.

Value applied: Apply the following default values for the different waste types j:

Year	W_j (waste amount) (t/x)
2012	7,440
2013	89,425
2014	91,250
2015	93,075
2016	94,900
2017	96,725
2018	98,915
2019	100,740
2020	102,930
2021	105,120
2022	106,945
2023	109,135
2024	111,325
2025	113,515
2026	115,705
2027	118,260
2028	120,450
2029	123,005
2030	125,560

Waste type of Landfill site	Weight content（%）
W_1-Wood and wood products	4.09
W_2-Pulp, paper and cardboard	8.32
W_3-Food, food waste, beverages and tobacco	42.16
W_4-Textiles	2.71
W_5-Garden, yard and park waste	12.14

W_6-Glass, plastic, metal other inert	*30.58*

Data / Parameter:	*$EF_{grid,OM,y}$*
Data unit:	*tCO_2/MWh*
Description:	*Operating margin CO_2 emission factor in year y*
Source of data used:	*2023 Baseline Emission Factors for Regional Power Grids in China dated 09/07/2024 published by China DNA*
Value applied:	*0.9014*

Data / Parameter:	*$EF_{grid,BM,y}$*
Data unit:	*tCO_2/MWh*
Description:	*Build margin CO_2 emission factor in year y*
Source of data used:	*2023 Baseline Emission Factors for Regional Power Grids in China dated 09/07/2024 published by China DNA*
Value applied:	*0.3597*

Data / Parameter:	*$EF_{grid,CM,y}$*
Data unit:	*tCO_2/MWh*
Description:	*Emission factor of CCPG*
Source of data used:	*2023 Baseline Emission Factors for Regional Power Grids in China dated 09/07/2024 published by China DNA*
Value applied:	*0.63055*

Data / Parameter:	*NCV_{CH4}*

Data unit:	*MJ/Nm³*
Description:	*Net calorific value of methane at reference conditions*
Source of data used:	*AMS-III.G*
Value applied:	*35.9*

Data / Parameter:	*EE_y*
Data unit:	*%*
Description:	*Energy Conversion Efficiency of the project equipment*
Source of data used:	*AMS-III.G*
Value applied:	*40*

<u>Parameters determined ex-post</u>

- *Policy requirement: $F_{CH4,BL,y}$, $\rho_{reg,y}$*

Parameters title	*Descriptions*
$F_{CH4,BL,y}$	*Amount of methane in the LFG which is flared due to a requirement in year y*
$\rho_{reg,y}$	*Fraction of LFG that is required to be flared due to a requirement in year y*

$F_{CH4,BL,y}$ and $\rho_{reg,y}$ will be checked annually through information of the host country's regulatory requirements relating to LFG, contractual requirements, or requirements to address safety and odour concerns. The process, measurement methods, procedures and calculation of $F_{CH4,BL,y}$ has been presented in the 3.3.6 of the report and the applied value is determined as 20% as case 2 was selected.

- *Electricity Generation: $EG_{PJ,facility,y}$, EG_y*

Parameters title	*Descriptions*
$EG_{PJ,facility,y}$	*Quantity of net electricity generated supplied by the project plant/unit in year y*
EG_y	*Electricity generation in year y*

$EG_{PJ,facility,y}$ and EG_y will be continuous measured by electricity meters with accuracy of 0.5s and monthly recorded. The electricity meters will be calibrated in accordance with national standards or other relevant requirements. The accuracy of electricity meters will be satisfied with relevant standards or requirements. The type, serial number, location, function and accuracy of the electricity meters have been confirmed through site visit.

Through the site visit, the validation team confirm the correctness of following information:

a) Methods for measuring, recording, storing, aggregating, collating and reporting data and parameters;

b) Policies for oversight and accountability;

c) Procedures for internal auditing and QA/QC;

d) Procedures for handling non-conformances with validated monitoring plan.

By site visit, it is confirmed that manufacturer details of the monitoring equipment presented in the PD is accurate and in line with the actual situation.

By site visit and checking related regulations, it is confirmed that the calibration period of the monitoring equipment presented in the PD is accurate and in line with the requirements in related regulations.

By site visit, it is confirmed that the training and maintenance procedures and emergency procedures for monitoring system is in place and in line with the description in the PD.

Based on the on-site visit and interviewed with the Manager of Suide County BCCY New Power CO., Ltd., the validation team confirmed that monitoring parameter has been correctly descried in the PD and in compliance with the methodology AMS-III.G Version 10.0, AMS-I.D Version 18.0 and the guidance given by EB via CDM Validation and Verification Standard for project activities version 03.0 and VCS via VCS Standard version 4.7. Also, the monitoring equipment and location has been confirmed through the site visit and are in line with the requirements of methodology.

In conclusion, the assessment team confirmed that all parameters those fixed at validation and those that need to be monitored in line with applied methodology are included in the monitoring plan and in line with methodology.

3.4 Non-Permanence Risk Analysis

Not applicable for the present project activity.

4 VALIDATION OPINION

4.1 Validation Summary

Applus+ Certification has been engaged by Henan BCCY Environmental Energy Co., Ltd. to perform the validation of Suide County Urban MSW Landfill Biogas Pollution Control and Comprehensive Utilization Project (VCS Ref. No. 5281).

The management of the project proponent/owner is responsible for the preparation of the GHG emissions data and the reported/estimated GHG emissions reductions on the basis set out within the project's Monitoring Plan and the approved methodology AMS-III.G, version 10.0 and AMS.I.D, version 18.0.

4.2 Validation Conclusion

Our Validation approach was based on the requirements as defined under the Kyoto Protocol, Marrakesh accord, as well as those defined by the CDM Executive Board and VCS board. Our approach is risk-based, drawing on an understanding of the risks associated with estimated GHG emissions data and the controls in place to mitigate these. The validation can confirm that:

The project's description compliance with, the requirements of Article 12 of the Kyoto Protocol, the CDM Modalities and Procedures as agreed in the Marrakech Accords under decision 3/CMP.1, the annexes to this decision, subsequent decisions and guidance made by COP/MOP & CDM Executive Board and other relevant rules, including the Host Country legislation and sustainability criteria along with VCS Program Guide version 4.4 and VCS standard version 4.7.

The project's baseline and additionality and monitoring plan are assessed against AMS-III.G, version 10.0 and AMS.I.D, version 18.0 for small scale project.

A risk based approach has been followed to perform this validation activity. The review of the project description and additional documents related to baseline and monitoring methodology; the subsequent background investigation, follow-up interviews with Project Owner have provided LGAI Technological Centre S.A. (Applus+ Certification) with sufficient evidence for positive validation opinion as per the requirement of VCS.

Crediting Period: From 24/02/2023 to 23/02/2033

Validated estimated GHG emission reductions and carbon dioxide removals for the project crediting period:

Vintage period	Estimated baseline emissions (tCO_2e)	Estimated project emissions (tCO_2e)	Estimated leakage emissions (tCO_2e)	Estimated reduction VCUs (tCO_2e)	Estimated removal VCUs (tCO_2e)	Estimated total VCUs (tCO_2e)

24/02/2023 – 31/12/2023	23,078	0	0	23,078	0	23,078
2024	29,216	0	0	29,216	0	29,216
2025	31,310	0	0	31,310	0	31,310
2026	33,370	0	0	33,370	0	33,370
2027	35,409	0	0	35,409	0	35,409
2028	37,418	0	0	37,418	0	37,418
2029	39,408	0	0	39,408	0	39,408
2030	41,381	0	0	41,381	0	41,381
2031	39,338	0	0	39,338	0	39,338
2032	37,399	0	0	37,399	0	37,399
01/01/2033 – 23/02/2033	5,261	0	0	5,261	0	5,261
Total	352,588	0	0	352,588	0	352,588

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification	Assessment method and conclusion
Not applicable	*Not applicable*	*Not applicable*	*Not applicable*

APPENDIX II: < REFERENCE LIST >

1. *PD version 01 dated 16/11/2024, version 03 dated 28/07/2025*
2. *ER calculation spreadsheet for ex-ante estimation*
3. *Board resolution dated 12/09/2022*
4. *Approved methodology AMS-III.G: Landfill methane recovery, version 10.0*

   Approved methodology AMS-I.D: Grid connected renewable electricity generation, version 18.0
5. *CDM Validation and Verification Standard for project activities, version 03.0*
6. *CDM Project Standard for project activities, version 03.0*
7. *Tool to calculate the emission factor for an electricity system, version 07.0*
8. *Emissions from solid waste disposal sites, version 08.1*
9. *IPCC2006 Guidelines for National Greenhouse Gas Inventories*
10. *Tool to determine the mass flow of a greenhouse gas in a gaseous stream, version 03.0*
11. *Positive lists of technologies, version 04.0*
12. *VCS standard version 4.7, dated 16/04/2024*
13. *VCS Program Guide version 4.4, dated 17/01/2023*
14. *Business License of Suide County BCCY New Power CO., Ltd.*
15. *Feasibility Study Report dated 12/2021*
16. *Project Approval dated 14/01/2022*
17. *Environmental Impact Assessment Report dated 07/2022*
18. *Environmental Impact Assessment (EIA) Approval dated 24/08/2022*
19. *Nameplate of the equipment*
20. *Construction Order dated 30/09/2022*
21. *Stakeholder Meeting Minutes dated 13/09/2022*
22. *Local stakeholder consultation questionnaire*
23. *Stakeholder Meeting Notice dated 29/08/2022*
24. *Grievance Expression Book located in the project site*
25. *Operation log covering the monitoring period*
26. *Cooperation agreement between Urban Management Law Enforcement Bureau of Suide County and Henan BCCY Environmental Energy Co., Ltd. dated 13/12/2021*

27. *2023 Baseline Emission Factors for Regional Power Grids in China dated 09/07/2024*

28. *Technical specification of equipment*

29. *Declaration for no double counting issued by project proponent*

30. *"Notice on Key Work Related to the Management of Greenhouse Gas Emission Reports for Enterprises in 2022" issued by Ministry of Ecology and Environment of P. R. China*

31. *REC Mechanism database of China: https://www.greenenergy.org.cn/*

32. *I-REC website: https://www.irecstandard.org*

Regulation and laws:

"Women's Rights Protection Law"

http://www.scio.gov.cn/ztk/xwfb/46/11/Document/976078/976078_3.htm

"Minor Protection Law"

33. *https://www.gov.cn/xinwen/2020-10/18/content_5552113.htm*

"Labor Law"

https://www.gov.cn/banshi/2005-05/25/content_905.htm

"Criminal Law"

http://gongbao.court.gov.cn/details/f8e30d0689b23f57bfc782d21035c3.html

34. *HR Records and Staff Roaster*

35. *VCS monitoring manual*

36. *Training Records covering this monitoring period*

37. *"Standard for Pollution Control on the Landfill Site of Municipal Solid Waste" (GB 16889-2008)*

38. *"Technical Code for Municipal Solid Waste Sanitary Landfill" (GB 50869-2013)*

APPENDIX III: <CLARIFICATION REQUESTS, CORRECTIVE ACTION REQUESTS, FORWARD ACTION REQUESTS (CAR/CL/FAR)>

CL ID	01	Section no.	3.4.5	**Date:** 16/11/2024
Description of CL				
Please supplement the timeline of the project in the 3.5 of the PD to demonstrate the history of the project				
Project proponent response				**Date:** 19/11/2024
The timeline of the project including major milestone such as project approval, EIA approval and etc. has been included in the updated PD.				
Documentation provided by project proponent				
Updated PD, FSR, EIA and etc.				
VVB assessment				**Date:** 20/11/2024
By site visit and checking supporting evidence, it is confirmed the timeline of the project including major milestone such as project approval, EIA approval and etc. has been included in the updated PD and confirmed to be correct. Therefore, CL01 was closed.				